

Pixels-to-Predictions: SmolVLM-500M QLoRA Fine-Tuning for ScienceQA

Yikai Sun

NYU Tandon School of Engineering

ys4035@nyu.edu

Abstract

We fine-tune HuggingFaceTB/SmolVLM-500M-Instruct for the *Pixels-to-Predictions* multimodal multiple-choice science QA competition under a strict 5,000,000-trainable-parameter budget and an offline-evaluation rule. We apply QLoRA (4-bit NF4 + double quantization) with rank-8 LoRA across all seven text-decoder linear projections, yielding 4,341,760 trainable parameters (87% of the cap). We score by single-token log-likelihood at the answer-letter position and apply a position-bias-aware test-time augmentation (smart-TTA) that averages logits over $K = 4$ random choice permutations while bypassing TTA on 2-choice questions where it empirically degrades accuracy. Our final submission — a logit-averaged ensemble of two adapters (standard LoRA and DoRA, both 5-epoch $r=8$) — achieves **0.81690** on the public leaderboard, +5.23 pp over the same model with standard scoring (0.76458). Six single-adapter ablations and one ensemble isolate which axes of capacity matter at this scale: training duration is the dominant lever (+3.6 pp from 3 to 5 epochs), but extending to 7 epochs *reduces* LB by 1.0 pp (overfitting), establishing 5 epochs as near-optimal; doubling LoRA rank from 4 to 8 contributes +1.8 pp at matched training duration; higher visual resolution, DoRA, and training-time choice-order augmentation are each statistically tied with baseline.

1 Introduction

The competition task is multimodal multiple-choice science QA: each example pairs an image (diagrams, maps, charts, photos) with a question, 2–5 textual answers, and optional hint / lecture context. The dataset matches the ScienceQA schema.

The dataset’s `num_choices` distribution is approximately 21 / 50 / 25 / 4% over {2, 3, 4, 5}, giving a random-guess baseline of $\sim 32.5\%$, *not* 25%. The base SmolVLM-500M, used zero-shot, achieves **41.5%** on a 200-example val sample (binomial SE ≈ 3.5 pp), with a strong “A” bias (71.5% of predictions choose the first option) and 5-choice accuracy below random (18.6%) — we hypothesize because the letter “E” rarely appears as a multiple-choice option in pretraining text. These two patterns motivate the loss-masked SFT and choice-permutation TTA below.

Contributions. (i) A QLoRA fine-tuning pipeline targeting all seven text-decoder projec-

tions at rank 8 (4.34M trainable, 87% of cap). (ii) A position-bias-aware smart-TTA that averages over $K = 4$ random choice permutations on 3+ choice questions, bypassing TTA on 2-choice. (iii) A data-curve-driven ablation methodology with the headline finding that training duration peaks at 5 epochs and overfits at 7, while five other axes (rank, image res, DoRA, choice-aug, ensemble lift) are statistically tied with baseline. (iv) Diagnosis and resolution of a QLoRA loss-explosion failure on T4 via paged AdamW 8-bit, grad clipping at 0.3, and warmup ratio 0.1.

Code: <https://github.com/Palomarr/pixels-to-predictions>.

2 Methodology

2.1 Architecture and Parameter Budget

SmolVLM-500M-Instruct consists of a frozen SigLIP-Base vision encoder (~ 93 M params), a frozen modality projector, and a SmolLM2-360M

text decoder (32 layers, hidden 960, 15 attention heads, 5 KV heads, MLP intermediate 2,560). We adapt only the text decoder via LoRA on all seven linear projections per layer (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj) at $r = 8$, $\alpha = 16$, yielding **4,341,760 trainable parameters** (87% of the 5M cap). Considered alternatives within the cap (attention-only at $r = 16$: 3.28M; MLP-only at $r = 8$: 2.70M) all sacrifice either MLP or attention adaptation; the QLoRA paper finds low-rank adaptation across all linear layers beats higher-rank attention-only at matched budgets. A regex `target_modules` constrains LoRA to text-decoder layers only — naive name-based matching would also hit the vision encoder’s same-named projections, wasting ~ 30% of the budget on frozen vision features.

2.2 Quantization: QLoRA

We load the base in 4-bit NF4 with double quantization (~ 250 MB VRAM vs ~ 1.0 GB at bf16); compute dtype is nominally bf16 but on T4 (sm_75, no native bf16 tensor cores) falls back to emulated kernels. The submission notebook runs offline without `bitsandbytes`, so it loads the base in full bf16 — introducing a precision-mismatch gap discussed in Section 6.1.

2.3 Prompt Format and Loss Masking

We follow the competition starter notebook’s template:

```
<image>
Context:
{lecture, truncated to 800 chars}
{hint}

Question: {question}
Choices:
  A. {choice_0}
  B. {choice_1}
  ...
Answer: {letter - only at training}
```

The `<image>` token marks where the processor splices in visual tokens. We force `do_image_splitting=False` and `longest_edge=384` for exactly 81 visual tokens per image; the processor’s default multi-tile splitting up to 1,536 px (~ 1,088 visual tokens) would blow past our 1,024-token sequence limit and

waste compute on most ScienceQA images, which are < 512 px natively.

Loss masking. During SFT we set labels to -100 at every token position *except* the trailing answer-letter token, concentrating 100% of the gradient signal on “predict the right letter” rather than on language-modeling the lecture/question/hint context. Without this, our 4.34M-parameter budget would be largely consumed by a training objective that has no test-time analogue.

2.4 Inference: Log-Likelihood Scoring

For each choice $j \in \{0, \dots, \text{num_choices} - 1\}$, we compute $\text{score}[j] = \log P(\text{letter}_j \mid \text{image}, \text{prompt})$ from the next-token logits at the answer position, where letter_j is the single token for the j -th letter (“ A”, “ B”, ...; the leading space matters under SmolLM2’s BPE). Letters beyond `num_choices` are masked to $-\infty$ before `argmax`. One forward pass per example, fully deterministic — avoiding the brittleness of generation parsing (variable formats, hallucinated letters).

2.5 Test-Time Augmentation: Smart Choice-Permutation

Pretrained-baseline analysis revealed strong position bias (A predicted 71% of the time); after fine-tuning, residual bias scales with the number of letter positions involved. For each test example we run $K = 4$ forward passes (pass 0 = identity, passes 1–3 = random shuffles), map each pass’s letter-position logit back to its original-choice slot via `accum[i, original_choice] += letter_logits[i, shuffled_position]`, and `argmax` the accumulated logits. Position bias cancels because each original choice appears in every position roughly evenly across passes.

Smart TTA: skip 2-choice. $K = 4$ TTA hurts 2-choice questions by -1.23 pp on val while helping 3-, 4-, 5-choice by $+1.18$, $+2.78$, $+4.55$ pp. With only two unique permutations available at $n = 2$, random sampling adds variance without canceling bias. A `tta_skip_below=3` parameter bypasses TTA below threshold, falling back to identity-pass. Smart-TTA yields $+1.91$ pp val and $+1.21$ pp LB on v1 — both larger than naive always-TTA.

2.6 Training Stability Fixes

Our first training run failed catastrophically: at step 20 the loss spiked to 2.66×10^{16} , and from steps 40–100 reported 0.000000 (gradient underflow). The pattern — 16 steps of stable forward+backward (one gradient-accumulation cycle) followed by a single divergent optimizer update — pointed to three independent fp16 issues that compound under QLoRA.

Fix	What it addresses
<code>optim="paged_adamw_8bit"</code>	Default <code>adamw_torch</code> keeps optimizer states in fp32; under QLoRA + bnb mixed-precision dequant, scale mismatches amplify. The 8-bit paged optimizer is the reference QLoRA recipe.
<code>max_grad_norm=0.3</code> (was 1.0)	Default 1.0 is calibrated for full fine-tuning. With only 4.34M params receiving gradients flowing through 511M frozen-quantized params, per-parameter step sizes are effectively much higher; aggressive clipping is the standard QLoRA practice.
<code>warmup_ratio=0.1</code> (was 0.03)	At 0.03, only ~ 12 warmup steps preceded peak LR= $2e-4$; the first optimizer update fired at step 16 with LR essentially at peak. 0.1 gives ~ 58 warmup steps, so by step 16 LR is $\sim 40\%$ of peak.

Table 1: QLoRA stability fixes that turned a divergent first run into reproducible training.

After applying all three fixes, training was stable across all subsequent runs.

3 Experimental Setup

Train 3,109 / val 1,048 / test 1,008 (labels held out). At effective batch 16 and 3 epochs, only ~ 580 optimizer steps. Paged AdamW 8-bit at LR 2×10^{-4} , cosine decay with 10% warmup, max grad norm 0.3, weight decay 0. Effective batch 16 (per-device $1 \times$ gradient accumulation 16); nominal bf16 (T4 fallback). `max_seq_length=1024`, lectures pre-truncated to 800 characters, images forced to a single 384×384 tile (~ 81 visual tokens). Single random seed (42). NVIDIA T4 (16 GB) on Kaggle Free Tier; 3-epoch run $\sim 2:44$, 5-epoch $\sim 4:50$, 7-epoch $\sim 6:30$.

4 Main Result

Our best submission applies the v3 adapter ($r = 8$, 384 px, 5 epochs) with smart-TTA at $K = 4$:

Configuration	Public LB
Random baseline (analytic)	~ 0.325
Pretrained, no fine-tune (200-ex val proxy)	~ 0.415
v1 fine-tuned, no TTA	0.76458
v1 fine-tuned + smart-TTA	0.77665
v2 (img 512) + smart-TTA	0.77867
v5 ($r = 4$, 5 epochs) + smart-TTA	0.79476
v4 (7 epochs) + smart-TTA	0.80281
v3 ($r = 8$, 5 epochs) + smart-TTA	0.81287
v6 (DoRA, 5 epochs) + smart-TTA	0.81488
v3 + v6 ensemble + smart-TTA	0.81690

Table 2: Public leaderboard scores. The pretrained-baseline row is a 200-example val proxy and not directly comparable to the LB rows (1,008-example hidden test); it anchors the qualitative gap from no-fine-tune to fine-tuned.

Net lift from training-side improvements (v1 $\rightarrow$ v6): +5.03 pp. Net lift from inference-only smart-TTA on v1: +1.21 pp. The v3+v6 ensemble adds another +0.20 pp (within one standard error of v6 alone) and is selected as the final submission as the highest measured LB.

5 Ablations

We vary training duration, LoRA rank, image resolution, adapter variant (LoRA vs DoRA), training-time choice-order augmentation, TTA strategy, and adapter ensembling. All runs use a single random seed; with $n = 1\{, \}$ 008 test examples and accuracies in 0.77–0.81, the binomial standard error of any single-run LB estimate is ≈ 1.3 pp, so we treat deltas < 2 pp as suggestive rather than significant.

5.1 Headline Ablations Table

ID	Config	Train-val	LB
v1	$r = 8$, all-7, 384 px, 3 epochs (baseline)	0.7634	0.77665
v2	1. img_size=512	0.7586	0.77867
v3	1. epochs=5	0.8015	0.81287
v4	1. epochs=7	0.7996	0.80281
v5	$r = 4$, $\alpha = 8$, 5 epochs	0.7729	0.79476
v6	DoRA, $r = 8$, 5 epochs	0.7948	0.81488
v7	1. choice-order training aug	0.7634	n/s
ens.	v3 + v6 logit-averaged	—	0.81690

Table 3: Train-val is 4-bit eval (training-time); LB uses bfl6 + smart-TTA. v1 LB without TTA: 0.76458. v6 trainable params: 4.62M (vs 4.34M for v1–v5, v7). v7 not submitted (“n/s”) because train-val identity to v1 predicts LB ≈ 0.78 , well below the v3+v6 ensemble.

5.2 TTA Strategy (No Retraining)

On v1’s val ($n = 1048$): no TTA = 0.7290, always-TTA ($K = 4$) = 0.7405 (+1.15 pp), **smart-TTA** ($K = 4$, skip 2-choice) = **0.7481** (+1.91 pp). Per-bucket smart-TTA: 2-choice 0.7992 (bypassed, identical to no-TTA), 3-choice 0.7106 (+1.18 pp), 4-choice 0.8492 (+2.78 pp), 5-choice 0.3182 (+4.55 pp). Position bias is largest where invalid letter slots are most numerous — TTA’s logit averaging cancels exactly that bias. Smart-TTA dominates, used for all submissions from v1+sTTA onward.

5.3 Image Resolution (v2)

The hypothesis was that ScienceQA’s diagram-heavy questions would benefit from higher visual token resolution. Empirically, train-val *fell* by 0.48 pp (0.7634 $\rightarrow$ 0.7586) while LB *rose* by 0.20 pp (0.77665 $\rightarrow$ 0.77867); both deltas within one standard error. We report this as **statistically tied with baseline**. Plausible reasons: ScienceQA images are mostly < 512 px natively (upsampling adds no information); SmolVLM was pretrained at specific image scales (512-px tiles are mildly OOD); at our parameter budget, visual fidelity is not the binding constraint.

5.4 LoRA Rank (v5) and Decomposition

The capacity-halving control (v5: $r = 4$, $\alpha = 8$, 5 epochs, ~ 2.17 M trainable params) yields **0.79476** LB. Crossing v1, v3, v5 isolates two axes at fixed everything-else: $r = 4 \rightarrow r = 8$ at epochs=5 contributes +1.81 pp (v5 $\rightarrow$ v3); 3 $\rightarrow$ 5 epochs at $r = 8$ contributes +3.62 pp (v1 $\rightarrow$ v3). Training duration is roughly $2 \times$ the lever of rank, indicating fitting time is a stronger constraint than fitting capacity at our parameter budget and dataset size.

5.5 Training Duration: The Convergence Curve

The single most impactful axis. Three measurements at epochs $\in \{3, 5, 7\}$ trace the generalization curve cleanly:

Epochs	Run	Train-val	LB
3	v1	0.7634	0.77665
5	v3 (best)	0.8015	0.81287
7	v4	0.7996	0.80281

Table 4: Convergence curve at $r = 8$. Both metrics peak at 5 epochs and decline at 7.

The v3 ablation was motivated by v1’s training loss — *still descending* at step 580 (final epoch-3 step, loss 0.2535; descent: 1.386 $\rightarrow$ 0.674 $\rightarrow$ 0.519 $\rightarrow$ 0.323 $\rightarrow$ 0.254) with no plateau — which we interpreted as under-training (9,300 example-views was insufficient). v3’s +3.62 pp LB lift confirmed this. The v4 result (5 $\rightarrow$ 7 epochs) is a textbook overfitting signature: both metrics decline, asymmetrically (-0.19 pp on train-val vs -1.01 pp on LB), consistent with fitting training-specific patterns that transfer to similarly-distrib-

uted val but not held-out test. Across the explored range, training duration affects LB by ~ 5 pp — larger than rank (~ 1.8 pp) or visual resolution (~ 0 pp).

5.6 DoRA: Adapter-Variant Control (v6)

DoRA (Liu et al., 2024) replaces LoRA’s additive update with a magnitude-direction decomposition $W' = m \cdot (W + \Delta W) / \|W + \Delta W\|$, adding one learned magnitude scalar per output channel (+277{,}480 params; 4.62M total trainable). v6 (DoRA on the v3 config) yields **0.7948** train-val and **0.81488** LB — statistically tied with v3 (Δ within one SE). DoRA produced no measurable lift over standard LoRA at this dataset and parameter scale; the val-LB inversion (train-val below v3, LB above) mirrors v1 $\rightarrow$ v2 and is discussed in Section 6.2.

5.7 Choice-Order Training Augmentation (v7)

v7 (v3 config + per-__getitem__ random permutation of choices, with answer-letter remapped) yields train-val **0.7634** — numerically identical to v1’s 0.7634, despite v7 training for 67% more steps. The augmentation transformed the SFT objective from position-conditioned (“for this image+question, predict letter B”) to content-conditioned (“predict the letter whose choice text matches the question”); at our 4.34M-parameter budget on 3,109 training examples, the harder content-conditioned objective regressed to v1’s plateau. **Position bias is therefore better addressed at test time via smart-TTA than at training time via augmentation**, at this scale. We did not submit v7 to the leaderboard.

5.8 Adapter Ensembling

A logit-averaged ensemble of v3 and v6 yields **0.81690** LB — +0.20 pp over v6’s 0.81488. The two adapters share architecture, training data, and seed, differing only by adapter variant (standard LoRA vs DoRA), so error correlation is high; the ensemble’s modest lift is consistent with that prior. The ensemble is our final submission.

6 Discussion

6.1 The 4-bit-vs-bf16 Inference Gap

Training-time val uses the 4-bit base; inference val uses bf16 (no bitsandbytes available offline). On v1 we measured this gap explicitly: train-val (4-bit) 0.7634 vs inference val (bf16, no TTA) 0.7290 — a **3.44 pp drop** from precision mismatch, larger than the < 1 pp typically reported in the QLoRA paper. Our LoRA-on-all-7-projections config (including high-magnitude MLP weights) is plausibly more sensitive to dequantization noise than attention-only. Bundling bitsandbytes wheels to enable 4-bit inference was disproportionate engineering risk for the expected 1–3 pp lift.

6.2 Val-vs-Leaderboard Ranking Flips

Two val-LB ranking inversions of similar magnitude were observed: v1 $\rightarrow$ v2 (train-val -0.48 pp, LB $+0.20$ pp) and v3 $\rightarrow$ v6 (train-val -0.67 pp, LB $+0.20$ pp). Each individual delta is within one SE, but the pattern repeating across two independently-trained pairs suggests either (i) val and test are not perfectly i.i.d. by per-example difficulty, so configs handling hard examples differently rank-flip; or (ii) the 4-bit $\rightarrow$ bf16 transition is non-uniform, with adapter variants (especially DoRA’s magnitude scalars) absorbing dequantization noise differently. We cannot disambiguate from single-seed measurements.

7 Reproducibility

Each of the seven reported runs (v1–v7) is registered as a named profile in config.ABLATIONS; one command reproduces any row of Table 3: `python train.py --ablation v3_epochs5`. Each completed run writes val_results.json with resolved config and val accuracy, pairing recipe with measurement. Submission scoring at `make_submission.py --tta_k 4 --tta_skip_below 3 --adapter_path A B` reproduces the smart-TTA + ensemble pipeline of our final submission. The repo’s README documents the competition-data staging, transformers-v5 import fallback, CUDA-state cleanup, and QLoRA stability fixes (Table 1) handled in code.

8 Conclusion

Under a fixed model, a 5M-trainable-parameter cap, and offline evaluation, we achieved **0.81690** on the public leaderboard via a logit-averaged ensemble of two QLoRA adapters trained at rank 8 across all seven text-decoder linear projections for 5 epochs (one with standard LoRA, one with DoRA), augmented with smart choice-permutation TTA. Training duration was the dominant lever by a wide margin (3 $\rightarrow$ 5: +3.62 pp; 5 $\rightarrow$ 7: -1.01 pp), bounding the optimum near 5 epochs. Higher visual resolution was statistically tied with baseline; halving the rank cost 1.81 pp; DoRA contributed +0.20 pp; choice-order training augmentation regressed to baseline. Across all five non-training axes we tested, no single architectural or augmentation lever moved LB by more than 2 pp, indicating that fitting time on the small training set — not capacity, adapter sophistication, or position-bias mitigation at training time — was the binding constraint. The 4-bit-vs-bf16 precision gap forced by the offline-eval rule accounts for ~ 3.4 pp of headroom otherwise available. All measurements use a single random seed ($SE \approx 1.3$ pp).